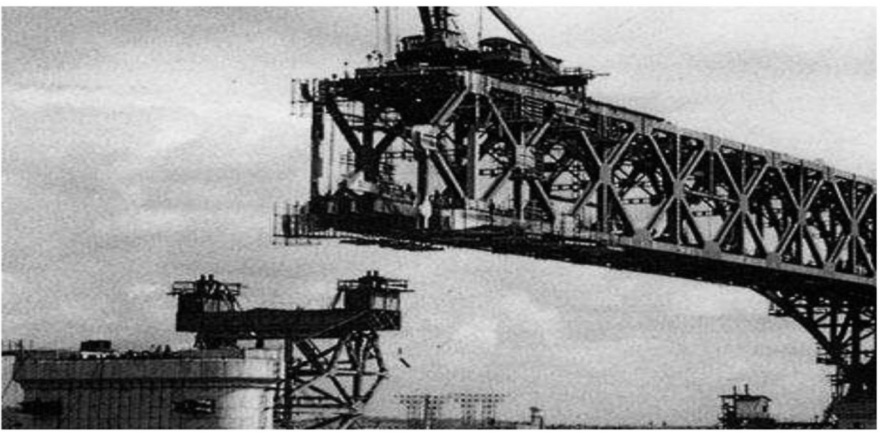


# 桥　梁

## 武汉建筑研究

中国是桥文化的故乡。自古就有“桥的国度”之称，发展于隋，兴盛于宋。遍布在神州大地的桥，有着悠久的历史，从桥梁的建筑设计折射出人类的科学理念以及情感意识



## Wuhan Architecture Research Bridges

贯穿武汉这座城的，不仅仅是流淌的江水更有那些架设在水面之上的桥梁，连接着城市的历史与未来，经济与文化，人与自然



“万里长江第一桥，两岸融云天际消”

# CONTENTS



## THREE BRIDGE

overview

## THREE BRIDGE

design





## THREE BRIDGE

culture

# Preface

Guardrail decoration on both sides of Wuhan Yangtze River Bridge

All the cases use figurative flowers, birds, insects, fish, animals and plants as modeling materials, vivid and lifelike images, with the most national characteristics of hollow paper-cutting techniques to shape, forming a beautiful and harmonious cast iron art.

There are two luxurious and elegant courtyard bridgeheads at each end of the bridge, both sides are a profound meaning of paper-cut carved patterns, each paper-cut carved patterns between the horizontal diamond-shaped Xiangyun pattern spacing, diamond-shaped Xiangyun map inlaid in the middle of the column, each column has a small lantern in the middle of the decoration, the whole constitutes a beautiful and harmonious Chinese arts and crafts long roll, Give people infinite beauty and imagination.

Wuhan is also a "river city", where there is a river, there will be a bridge, and Wuhan has a deep bond with the bridge.The Yangtze River flows in the heart of the city in the memory of sunset, we walk by the river.
See the sunset on the water,So slowly stroll to the Chinese bridge.Bridges run across the river,
From one end of the river to the other from ancient times to modern times.Ancient and modern blend, history and modern meet here, the rise of the city, everything bears the mark of the years.



01

# WUHAN THREE BRIDGES OVERVIEW

# 01 /
# Wuhan Three bridges Overview

## Wuhan Yangtze River Bridge

**Time**

1955.9-1957.9.25
The Yangtze River Bridge is the first large-scale railway and highway double-deck bridge after the founding of the People's Republic of China

**Space**: Starting from the south slope of Guishan in Hanyang and ending in the southeast at the top of Wuchang Snake Mountain into the river. Because the turtles and snakes lock the river, the river is narrow, shortening the length of the bridge. The bottom of the river is the residual of two mountains, except for the poor geological condition of pier No. 7, the seven piers of the bridge are standing on solid rock. Using the mountain conditions on both sides of the bridge, the clearance height under the bridge can meet the needs of navigation.

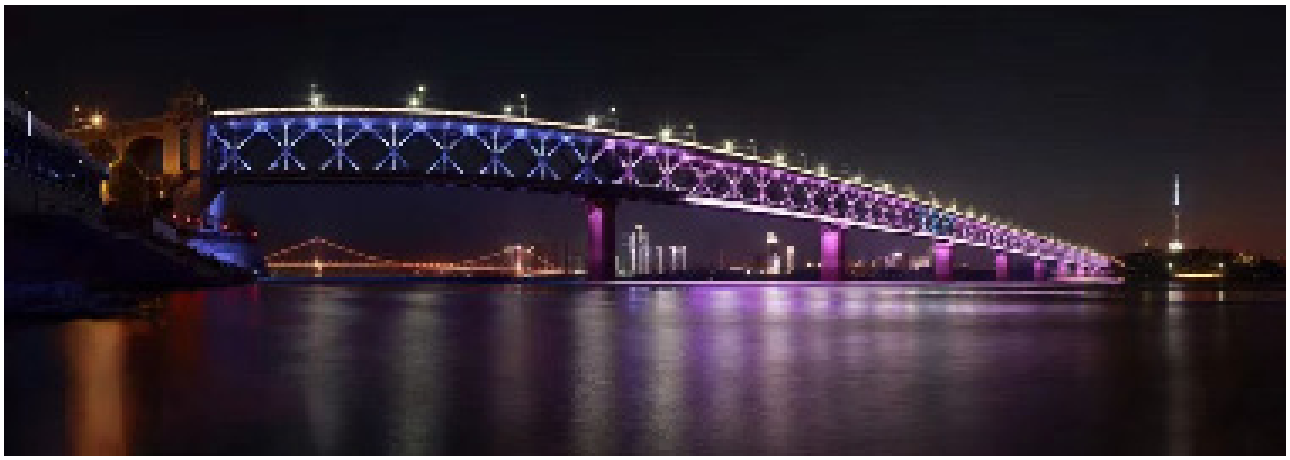
**Author**: Peng Min led a delegation from the former Chinese Ministry of Railways to Moscow with all the design drawings and technical materials of Wuhan Yangtze River Bridge to ask for the help



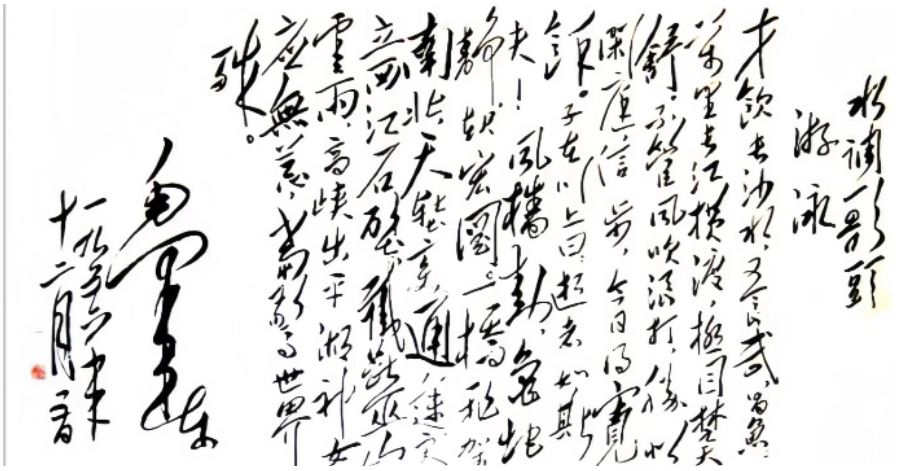
bridge experts to repeatedly study the Wuhan Yangtze River Bridge plan. Later, the Soviet side sent bridge experts to Wuhan to provide technical guidance, and MAO Yisheng was responsible for it. He accepted the task of building China's first bridge across the Yangtze River, the Wuhan Yangtze River Bridge. The bridge was completed in three years.

**Location**: Between Snake Mountain and Guishan, Hanyang, Wuchang District, Wuhan City, Hubei Province.
Lights on: 9:30 p.m. to 22 p.m









**Artist**: Chinese President writes poem on Yangtze River Bridge."Water song head • Swimming" wrote in "the wind is moving, the turtle snake is quiet, the grand plan." A bridge flies north and south, and the natural gap becomes a thoroughgoing road."

**Artwork**: Colorful Wuhan Yangtze River Bridge by Xiao Feng
Oil on canvas 120×150cm
The Last Steel Beam by Takeshi
81cm×53cm 1957

**Stamps related to the Yangtze River Bridge**

On October 1, 1957, the Chinese People's Post Office issued a set of 2 commemorative stamps of Ji 43 "Wuhan Yangtze River Bridge".

On October 1, 1959, the Chinese People's Post Office issued Ji 69 "The 10th Anniversary of the Founding of the People's Republic of China" commemorative stamp, of which the main picture of the fourth stamp is the Wuhan Yangtze River Bridge.
On July 16, 1976, the Chinese People's Post Office issued the J10 ("J" stands for commemorative stamp) commemorative stamp "To the Great River and the Sea to Exercise", of which the second stamp is also set against the background of the Wuhan Yangtze River Bridge.






# Wuhan Gutian Bridge

**Time**: :2007.8-2009

**Space**: Wuhan Jianghan No. 6 Bridge Project is located between Jianghan No. 5 Bridge and Jianghan No. 2 Bridge of the Third Ring Road, and the Han Port connects with Gutian No. 2 Road. The main bridge across the Han River is 472m long, and the standard section is 44m wide according to the plan, with 8 lanes in both directions. Two ramps are set up on the side of Hankou to connect with Yanjiang Avenue.
**Location:** Gutian Bridge is located between Jianghan No. 5 Bridge and Jianghan No. 2 Bridge on the Third Ring Line, and the Han Port connects with Gutian No. 2 Road.
**Author**: Lighting time: Starting at 18 p.m



# Parrot Chau Bridge



**Time**: :2007.8-2009
The Zhuzhou Yangtze River Bridge is located in Hanyang District, Wuhan City, Hubei Province, China, connecting Hanyang District and Wuchang District, is the eighth Yangtze River bridge in Wuhan City, and the world's first three-tower four-span suspension bridge with continuous main cable.

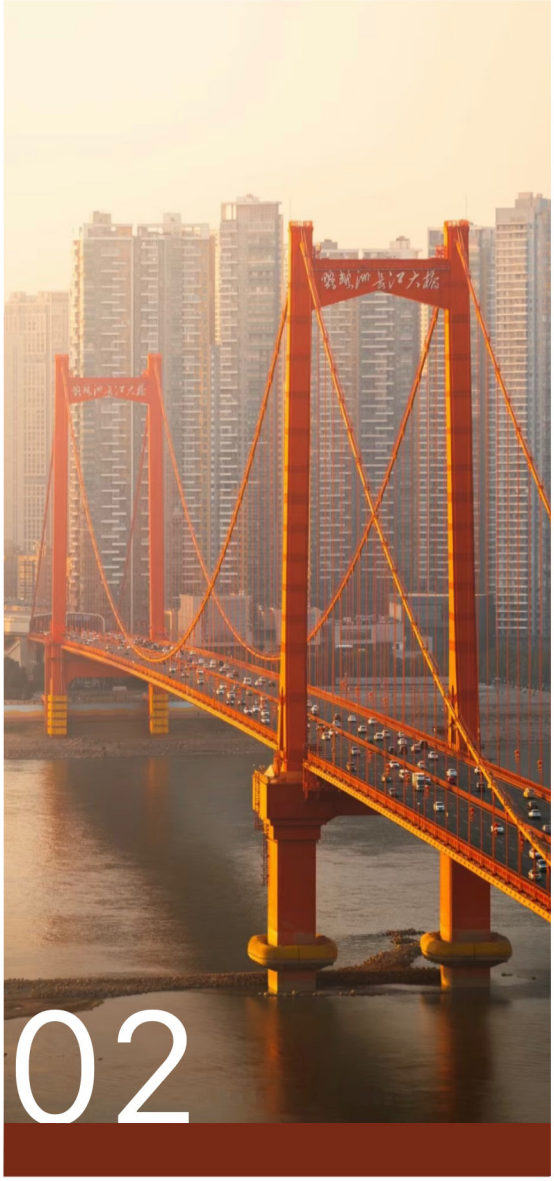
**Space**: The Zhuozhou Yangtze River Bridge is composed of three bridge towers, north and south, and each interchange ramp. The Hanyang connection project starts from Jiangcheng Avenue and connects to the main bridge of Zhuozhou Bridge along Mapu Road to ZhuoAvenue.

**Location**:  300 meters northwest of Shanhe Jiangjingge, Wuchang District, Wuhan City, Hubei Province.

**Lighting time**: 19.30 to 22.30 p.m.

**Author**: China Railway Bridge Bureau

**Artist**: Photographs by Zhang Xiang



02

# WUHAN THREE BRIDGES DESIGN

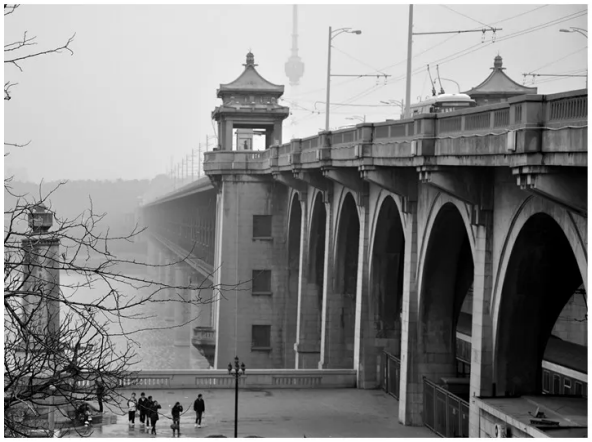


# 02 / Wuhan Three bridges design

## Bridge types

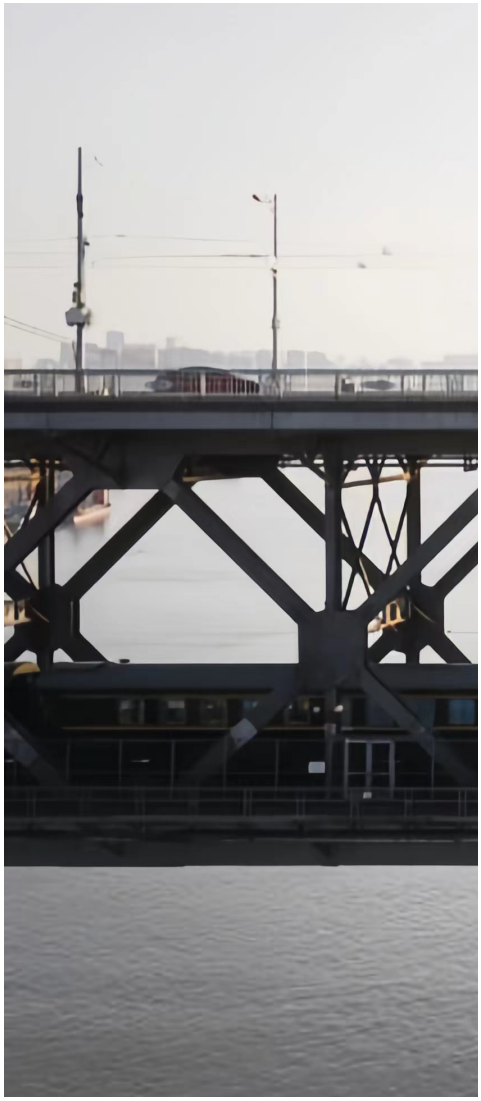
There are many types, which can be roughly divided into ancient Bridges, road Bridges, railway Bridges, overpasses and so on. Ancient Bridges are mainly made of wood and stone, such as the Zhuozhou Yangtze River Bridge on the south bank of the Yangtze River; Highway Bridges are mainly cross-river Bridges connecting various areas of the city, such as Wuhan Yangtze River Bridge; Gutian Bridge is a self-anchored suspension bridge.







# Yangtze River Bridge



**Design idea**

It was Zhang Zhidong, governor of Huguang in the late Qing Dynasty, who first proposed the plan of building a bridge across the river in Wuhan. From the perspective of architectural structure, the Yangtze River Bridge fully demonstrates the wisdom and courage of Chinese engineers, and its design and construction is undoubtedly an engineering miracle.

**Design features**

The main bridge of Wuhan Yangtze River Bridge is in the form of steel truss, and the approach bridge is in the form of arch. The steel frame is in sharp contrast in form and material. At the same time, arch structure is widely used in ancient Chinese Bridges, and the use of arch is easy to arouse people's sense of identity for such treatment. The bridge deck adopts the traditional Chinese double eaves, four slopes and




the middle of the column, each column has a small lantern in the middle of the decoration, the whole constitutes a beautiful and harmonious Chinese arts and crafts long roll, Give people infinite beauty and imagination.

**Design introduction**

The total length is 1670.4 meters, and the length of the main bridge is 1155.5 meters

**Design speed**

100 km/h for the upper deck and 160 km/h for the lower deck.
Design features: Double-deck passage, upper car, lower train, with pedestrian passage, geographical advantages, convenient transportation

**Design structure**

upperbridge panel, lower railway track, four bridgeheads and each interchange ramp. The calculated span of each hole is 128 meters.

# Wuhan Gutian Bridge

## Design idea

Gutian County as an important agricultural county, its traffic conditions lag behind, the government decided to build a bridge to improve the local traffic conditions

## Design features

Gutian Bridge is an ecological and environmental protection bridge. During the construction process, the designers paid attention to the concept of environmental protection and adopted many environmentally friendly materials and technologies. The bridge deck is made of permeable materials, which can effectively reduce the erosion and erosion of the rain on the bridge deck. At the same time, there are ecological wetlands under the bridge, providing a good ecological environment for aquatic organisms. Combining Chinese and Western elements, the bridge has a broad structure and adopts the architectural landscape design of the past and the present. The two main towers are designed into two grand arches on the bridge floor, and the triumphal arch of pale Cyrus blue is adopted.

In the center of the arch, there is a hollow Wuhan flower - plum blossom shape, showing the local cultural characteristics of Wuhan. Gutian Bridge is also a self-anchored suspension bridge with double tower and double cable plane, the main beam is a steel-concrete composite beam, the main beam structure is formed by paving the bridge panel on the steel beam, and the concrete box girder is added on both sides as the anchor span. The main cable is dispersed at the loose cable sleeve and anchored on the anchor surface behind the concrete box girder. The lower tower column of the main tower is a concrete tower column, the middle tower column is a lattice tower column composed of four tower limbs, and the top of the tower is a concrete tower top and a steel beam.

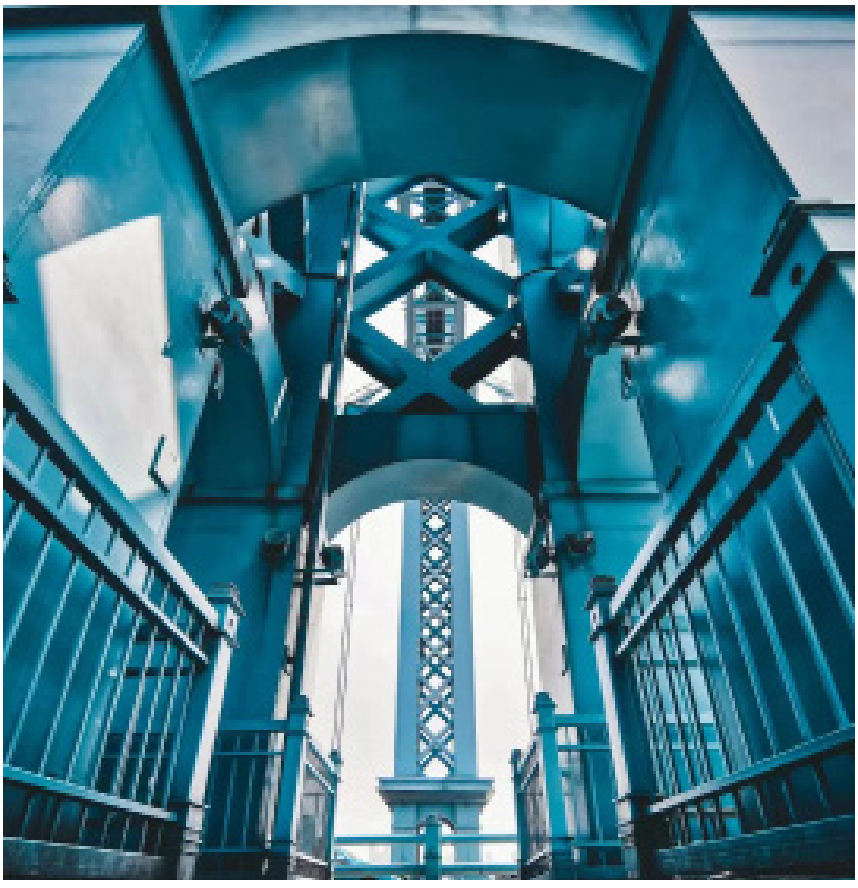
## Design introduction

The total length of the bridge line is 3050.1 meters, the total length of the main bridge is 2953 meters, and the bridge deck is designed as a two-way six-lane urban express road

## Design speed

60 km/h 123





# Zhuzhou Yangtze River Bridge

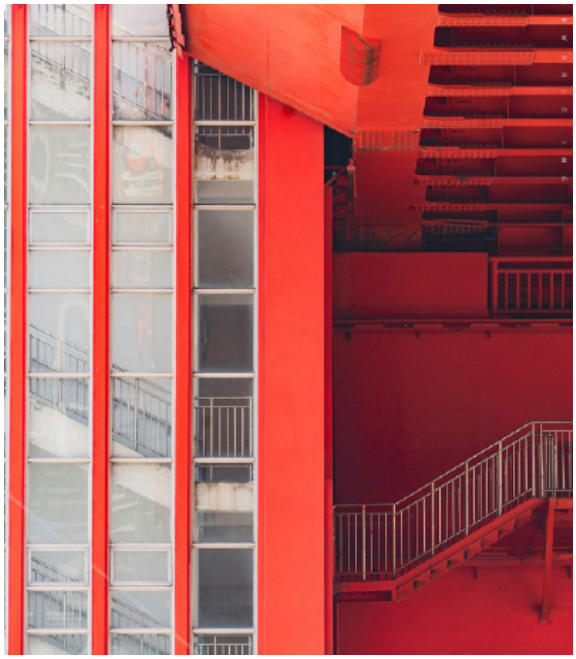
## Design idea

To ensure the safe and smooth passage of large ships on the Yangtze River

## Design features

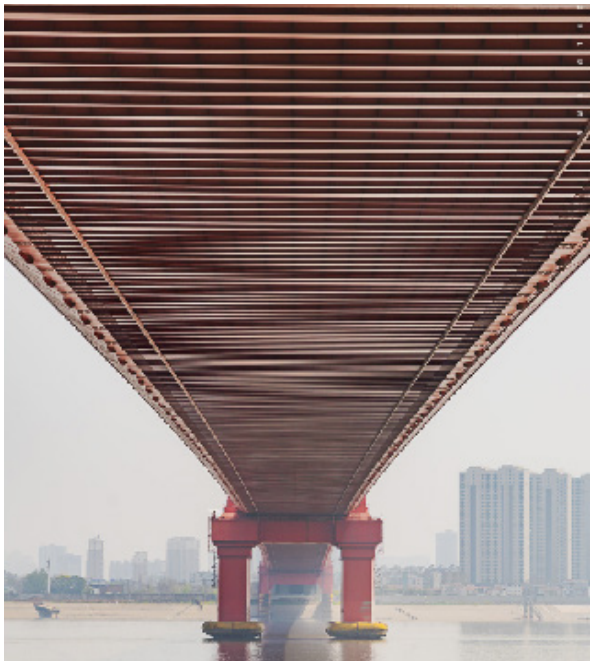
A combination of Chinese and Western architectural styles, it is an important transportation hub connecting the city and a landmark construction structure with cultural significance. The overall layout of the Zhuozhou Yangtze River Bridge is composed of three bridge towers, north and south, and each interchange ramp, and the main bridge section is arranged in the northwest to southeast direction.





## Design concept

Liuzhou Yangtze River Bridge is located less than 3 kilometers away from the upper reaches of Wuhan Yangtze River Bridge and within the sight line of Yellow Crane Tower. For the bridge type design, the designer gave up the choice of cable-stayed bridge to avoid the impact of nearly 200 meters high bridge towers on the urban landscape, and adopted the scheme of three-tower four-span steel-hybrid stiffened beam suspension bridge with moderate tower height. It is coordinated with the landscape of Wuhan Yangtze River Bridge.
The undulating shape of the main cable is rhythmic and beautiful, and the steady momentum of the bridge tower echoes the vast Yangtze River. The three towers stand tall, implying the comprehensive, coordinated, sustainable and balanced vigorous development of the three towns in Wuhan. In the painting color



of the Zhuozhou Yangtze River Bridge, in order to pursue a better landscape effect, the designer painted the bridge tower and other parts of the fashionable international "orange" red to coordinate with the steel beam, which can not only play a protective role, but also reflect the landscape effect after lighting.

## Design introduction

The total length is about 3.42 km 1234, the length of the main bridge is about 2.1 km 34, the width of the bridge floor is 38 meters 1234, the lane design is two-way 8 lanes 1234

## Design speed

60 km/h 24.
Investment scale: The total investment is about 5.029 billion yuan.



03

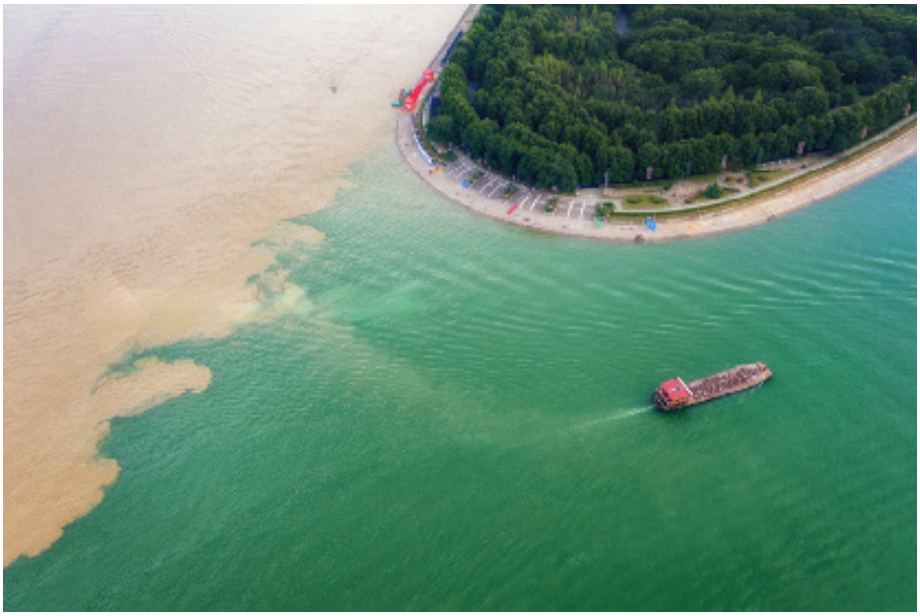
# WUHAN THREE BRIDGES CULTURE

# 03 /
# Wuhan bridge culture

## General overview

China is the hometown of bridge culture, known as "the country of Bridges" since ancient times, developed in Sui Dynasty and flourished in Song Dynasty. The bridge culture, which is woven into an extensive transportation network and connects the motherland in all directions, has a long history and rich cultural origins. The architectural design of Bridges reflects the scientific concept and emotional consciousness of human beings. A bridge building is like a large public sculpture, after completion, it integrates into the surrounding environment and forms a myriad of relationships. The bridge itself carries the change of historical background and time, so that we can see the precipitation of ethnic customs, religious beliefs, aesthetic tastes and other cultures.

Among the many cities in China, Wuhan stands out for its unique geographical location and rich cultural heritage.
The city, sandwiched between the Yangtze River and the Han River, has been known as the "Jianghan Dynasty Emperor" since ancient times.
And through the city, not only the flowing river, but also those Bridges erected on the water surface, they are like a fine link, connecting the city's history and future, economy and culture, people and nature.





# Yangtze River Bridge

01

its completion not only marks a major leap in Wuhan's transportation infrastructure, but also a milestone in China's modern bridge construction. The Yangtze River Bridge has now become a landmark in Wuhan, attracting countless tourists to stop and watch.

02

It bears witness to the important position of Wuhan in history, connecting the two cultural regions of Central Plains and Jiangnan. Not only that, the bridge also incorporates many traditional cultural elements, demonstrating the deep feelings of the Chinese people for the mother river of the Yangtze River. It is not only a miracle of traffic engineering, but also a symbol of Wuhan's culture and history. This building, known as "the first bridge of the Yangtze River", has always touched us with its magnificence and cultural connotation.

03

Since its completion in 1957, the Wuhan Yangtze River Bridge has accompanied the Chinese people for 66 years. As China's first independently designed and built public-rail bridge, it carries the memories and emotions of countless people and has now been listed as a key national cultural relic protection unit.





## Miracle on the Yangtze River

Since ancient times, the Yangtze River Rift has become a natural barrier between north and south, restricting people's life and trade. It was not until 1957 that the completion of the Wuhan Yangtze River Bridge completely changed this history. Its emergence not only opened up the north-south traffic artery, but also marked a huge leap forward in the field of infrastructure construction in New China.

## The bridge through the wind and rain

Since its completion, the Wuhan Yangtze River Bridge has undergone many large and small repairs and maintenance. Despite the passage of time, the bridge has maintained its unique appearance and solid structure. Today, the bridge has become one of the city's landmark buildings, witnessing the historical changes of Wuhan and even the whole of China.

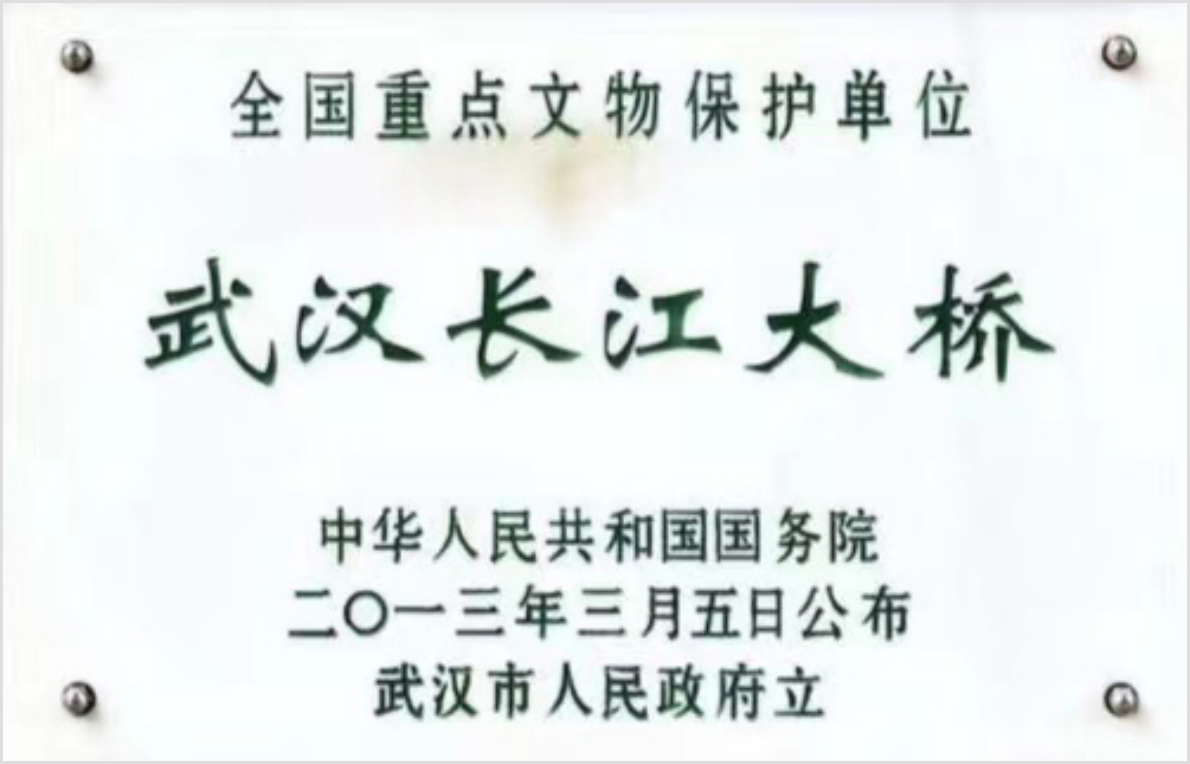


## Inheritance of culture

As a cultural heritage of high historical value, Wuhan Yangtze River Bridge was inscribed on the World Heritage List in 2007. Its existence not only shows the wisdom and courage of the modern Chinese people, but also becomes one of the important representatives of the national image.

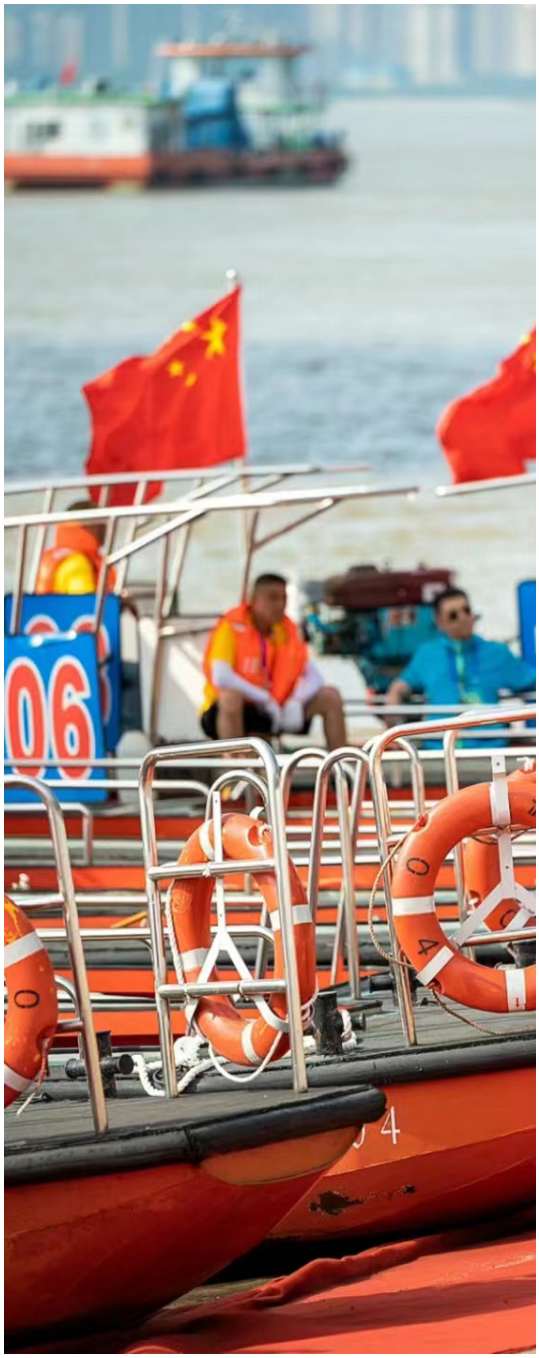
## Responsibility of cultural relics protection units

As a national key cultural relic protection unit, the protection of Wuhan Yangtze River Bridge has attracted much attention. The government and all sectors of society are actively participating in the protection and inheritance of the bridge to ensure that this historical treasure can be passed on forever.
Wuhan Yangtze River Bridge is not only a masterpiece in the history of Chinese bridge construction, but also a symbol of the Chinese nation from weak to strong. In the future, we will continue our efforts to protect this precious cultural heritage as an important link between the past, the present and the future. Let's witness the brilliant history of Wuhan Yangtze River Bridge and offer our best wishes for its future prosperity and inheritance.


全国重点文物保护单位
武汉长江大桥
中华人民共和国国务院
二〇一三年三月五日公布
武汉市人民政府立

# Explore more possibilities

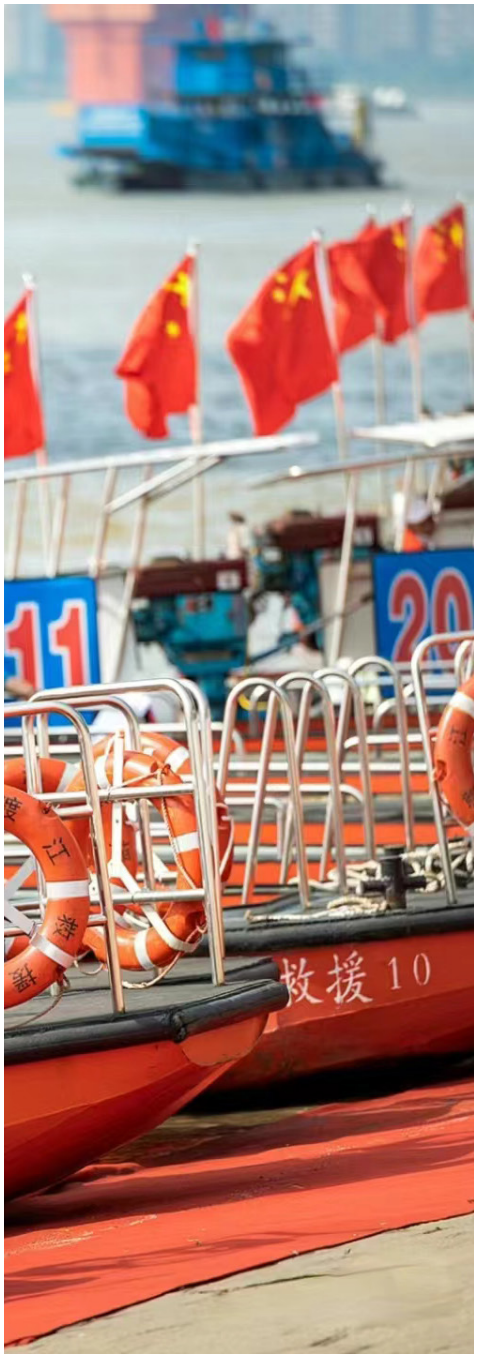
As a cultural relic protection unit, the cultural value of the Wuhan Yangtze River Bridge far exceeds its role as a transportation hub. However, the bridge's potential goes far beyond that. In the future, we can explore more ways to use this historical treasure to continue its role.




## For exploration

① Tourism development: As one of the world's famous Bridges, Wuhan Yangtze River Bridge can further develop its tourism resources. By rationally planning tourism routes and improving tourism supporting facilities, we will attract more tourists from home and abroad to come for sightseeing, tourism and pilgrimage. At the same time, the surrounding natural landscape and cultural attractions can be connected to form a distinctive tourist route, so that tourists can more comprehensively understand and feel the historical and cultural heritage of Wuhan.

② Cultural and creative industry: the cultural and creative industry with Wuhan Yangtze River Bridge as the theme also has great potential for development. It can combine modern scientific and technological means and creative design concepts to develop more cultural products, film and television works and cultural and creative derivatives with the characteristics of the bridge. At the same time, cultural and artistic activities and exhibitions with the theme of Wuhan Yangtze River Bridge can be held to attract more literary and artistic young people and creative talents to participate in them, and inject new vitality into the development of the city's cultural industry.




③ Environmental protection public welfare activities: As a national key cultural relic protection unit, Wuhan Yangtze River Bridge is also an important part of the cause of ecological environmental protection. Public awareness of environmental protection and cultural relic protection will be enhanced through environmental public welfare activities with the theme of protecting the Yangtze River and caring for cultural relics. At the same time, these activities can be used to raise funds for the maintenance of the bridge and related environmental protection projects, forming a virtuous circle of sustainable development.

④ Academic research: As a cultural heritage with high historical value, the Wuhan Yangtze River Bridge is also the focus of many scholars and researchers. By carrying out relevant academic research work, further excavating and sorting out the historical background, architectural technology and cultural connotation of the bridge and other aspects of information, so as to provide more abundant materials and data support for protection, inheritance and research.

# Wuhan Gutian Bridge



01

Multi-functional bridge of environmental protection culture. Gutian Bridge is not only a beautiful bridge, but also a multi-functional bridge integrating environmental protection, culture and interaction. In the construction process, pay attention to the concept of environmental protection, the use of water-permeable materials to reduce rain erosion. The bridge is carved with Wuhan's historical and cultural patterns, allowing visitors to enjoy the beautiful scenery and feel the city's deep historical heritage.

02

Gutian Bridge is a bridge with deep cultural heritage. On the bridge, visitors can enjoy many exquisite cultural elements. For example, there are historical and cultural patterns of Wuhan carved on the bridge, so that visitors can feel the deep historical and cultural heritage of Wuhan while enjoying the beautiful scenery.

03

Gutian Bridge is an interactive bridge. On the bridge, visitors can take photos with their families and friends and leave good memories. At the same time, there are also observation decks and rest areas on the bridge for tourists to rest and enjoy the scenery.

04

The blue romance of Gutian Bridge. Set foot on the Gutian Bridge, as if into a blue dream.The bridge, with its pale Prussian blue posture, stands above the Han River like a delicate ribbon, gently tied in the heart of the city. The blue theme of the bridge is not just for aesthetics. In Chinese culture, blue often symbolizes tranquility, depth and lofty ideals. The blue color of Gutian Bridge is like an attitude that the city of Wuhan shows to the world - it is calm and tolerant, and dares to dream. The blue of Gutian Bridge in the distance, let people feel quiet, in the near feeling, and let people infinite thoughts. On the bridge, you can see morning runners sweating, you can see couples holding hands and enjoying a good time, and you can also see photographers capturing





every moving moment of the city. Gutian Bridge hathus become an indispensable part of Wuhan people's lives. Not only that, the blue of Gutian Bridge will show different styles under different light and shadow changes. Whenever the sun sets, its blue figure becomes soft and deep under the sunset, whenever the night falls, the light will dress it up bright and dazzling, become a bright pearl on the Han River. People are here not only to cross the bridge, but also to feel the harmonious coexistence between the bridge and nature, the bridge and the city, and the bridge and people. The design inspiration of Gutian Bridge is said to come from the Arc de Triomphe in France. This design concept not only makes it different in shape, but also fits in with the culture of Wuhan city in spirit. Here, the wisdom of the East is intertwined with the romance of the West to create a new visual and sensory experience.



## 05

Standing on the Gutian Bridge, one side is the tenderness of the Han River, the other is the hustle and bustle of the city. The tall buildings in the distance and the waves and light of the river nearby form a picture of the combination of movement and movement.

Here, you can clearly see the development of Wuhan - the rise of high-tech industries and the protection of historical and cultural heritage complement each other. Wuhan, a city with both ancient and modern flavor, is taking Gutian Bridge as the hub to show its unique charm to the world. The experience of "appreciating the beauty of the city scenery along the Yangtze River in Wuhan" is not only a visual enjoyment, but also a spiritual touch. On the Gutian Bridge, visitors can not only witness the scenery of the city, but also feel the heartbeat of the city. As the famous saying goes, "The most beautiful scenery is the scenery in the mind." The beauty of Gutian Bridge and its surroundings makes people's hearts washed and sublimated. Walking through Gutian Bridge is not only a simple crossing, but also a spiritual journey.


BY WUWENPEI

# Zhuzhou Yangtze River Bridge



The Zhuzhou Yangtze River Bridge is not only a magnificent project, but also a fusion of history and culture. It was the imperial garden of Soochow during The Three Kingdoms Period and a military fortress during the Ming Dynasty. During the Qing Dynasty, it became an important port and trading center. Now, the completion of the Yangtze River Bridge not only facilitates the traffic between the two sides, but also becomes a new landmark building in Wuhan.

Near the Yangtze River Bridge in Zhuzhou, there are many food and cultural attractions. First, you can taste the local specialties - Wuhan hot and dry noodles and duck neck. In addition, you can also visit the Yellow Crane Tower, Yueyang Tower and other famous scenic spots. These attractions will give you an insight into the history and culture of China.There are many touching stories and interesting characters during the construction of the Zhuozhou Yangtze River Bridge. One of the most famous is the story of the family on and off the bridge. This story tells about an old man who patrols the bridge every day to protect his little grandson. The little grandson also plays under the bridge every day to accompany his grandfather. This story not only touched countless people, but also became a symbol of the Zhuzhou Yangtze River Bridge. Zhuozhou Yangtze River Bridge is a bridge full of history and culture. Here, you can feel the charm of nature and the warmth of humanity. At the same time, there are many food and attractions waiting for you to explore and discover.



“一桥飞架南北，天堑变通途”

